

Khandakar Rashedul Islam

Research Assistant | Environmental Researcher

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Summary

As an Environmental Science graduate, I have developed strong research expertise in microplastics pollution, water and wastewater treatment, environmental pollution, and biochar-based remediation strategies. I have contributed to several Q1 journal publications. Skilled in advanced analytical techniques, environmental sampling, and data analysis. Highly motivated to pursue master's or doctoral research aimed at developing innovative, science-driven solutions to global environmental challenges.

Research Interest

Emerging Contaminant (Microplastics, PFAS) | Water and Wastewater Treatment | Biochar | Heavy Metals

Academic Credentials

Master of Science (MS) in Environmental Science

Mar, 2023 – Dec, 2024

Jashore University of Science and Technology (JUST), Jashore-7408, Bangladesh

CGPA: 3.63 on a scale of 4.00

Thesis Title: Appraisal of microplastics identification, characterization and potential impacts in the Kuakata sea beach, Bangladesh

Bachelor of Science (B.Sc.) in Environmental Science

Jan, 2018 – Jan, 2023

Jashore University of Science and Technology (JUST), Jashore-7408, Bangladesh

CGPA: 3.39 on a scale of 4.00

Thesis Title: Performance evaluation of poultry litter and poultry litter biochar as a promising soil amendment for influencing agricultural soil properties and toxic metal availability.

Professional Experience

Research Assistant

Feb, 2022 – Present

Asia Arsenic Network (AAN)

Jashore-Benapole Highway, Pulerhat, Jashore-7400, Bangladesh

- Collected and analyzed environmental field samples across different matrices
- Prepared technical reports based on laboratory-generated data

Research Assistant

June, 2020 – Dec, 2024

Laboratory of Environmental Science and Technology

Jashore University of Science and Technology, Jashore-7408, Bangladesh

- Worked on multiple research projects and led five undergraduate student's thesis
- Designed experiments, conducted field sampling, managed lab work, analyzed data, and drafted manuscripts

Research Experience

Graduate Researcher – Microplastics Research

Jashore University of Science and Technology, Bangladesh | Feb 2023 – June 2024

- Explored microplastics pollution in different environmental matrices
- Led 5 undergraduate thesis students' research
- Applied FTIR, SEM, and chemical digestion protocols for MP identification

Worked as a research assistant in following projects:

2023 - 2024

Project Title: Microplastics in outdoor and indoor environments of urban area in Bangladesh: An emerging concern.

Funded By: Research cell, Jashore University of Science and Technology, Jashore 7408, Bangladesh.

Project Title: A first of its kind study on Microplastics pollution in surface water, sediment and soil in Bangladesh: occurrence, distribution, effects and potential solutions.

Funded By: Research cell, Jashore University of Science and Technology, Jashore 7408, Bangladesh.

Led the following research under my thesis supervisor:

Title: Microplastics in Agricultural Soil in Jashore, the Highest Vegetable Producing District of Bangladesh: Prevalence, Distribution and Characterization. (**Status:** Completed)

Title: Microplastics in Protected Pond and Harvested Rain Water: Identification, Characterization and Health Risk Assessment. (**Status:** Completed)

Title: Microplastics in Aquaculture Pond of Jashore, Bangladesh: Prevalence, Distribution, Characterization and Potential Risk Assessment. (**Status:** Completed)

Undergraduate Researcher – Environmental Research

Jashore University of Science and Technology, Bangladesh | Jun 2020 – Dec 2022

- Assessed biochar and bio-nano composites as soil amendments for environmental remediation
- Developed innovative approaches for water and wastewater treatment
- Utilized advanced analytical techniques (FTIR, AAS) for pollutant identification and characterization

Worked as a research assistant in following projects:

2021 – 2022

Project Title: Human health risk and hydro-geochemical appraisal of groundwater in the southwest part of Bangladesh by using GIS, water quality indices, and multivariate statistical approaches.

Funded By: Ministry of Science and Technology, Government of the people Republic of Bangladesh.

2019-2020

Project Title: Removal of toxic dye (methylene blue) from aqueous solution and industrial effluent by jute stick charcoal: adsorption isotherms and kinetics studies.

Funded By: Ministry of Science and Technology, Government of the people Republic of Bangladesh.

Worked as a research associate with thesis supervisor in following topics:

Title: Microplastics as an emerging pollutant in farmland soil: an impact evaluation study on farmland soil properties and toxic metal availability.

Title: Preparation, Characterization and Performance Evaluation of Bio-nano Composite for Sustainable Nutrients Release: Toward Sustainable Agricultural Practices.

Research Publications

Islam, K. R., Debnath, P. C., et al. (2025). Tracing the Journey of Microplastics in the Northern Bay of Bengal Marine Ecosystems, Bangladesh: A Comprehensive Study on Source Appointments, Distribution, and Potential Risks. **Journal of Hazardous Materials Advances (Elsevier, Q1, IF=7.7)**

Chowdhury, P., **Islam, K. R., et al. (2026).** Tracing Microplastic Footprints in Bangladesh's Largest Baors (Oxbow Lake): First Insights into Characterization, Ecological Risks, and Implications for Aquatic

Health. **Ecotoxicology and Environmental Safety (Elsevier, Q1, IF=6.1)**

- Biswas A., Debnath, P. C., **Islam, K. R.**, et al. (2026). Source Tracking, Pollution Load, and Potential Risks Assessment for Microplastic Pollution in Agricultural Soils, Bangladesh using Machine Learning and Multi-Matrix approaches. **Environmental Pollution (Elsevier, Q1, IF=7.3)**
- Rayhan, M. A., Debnath, P. C., Kharel, G., **Islam, K. R.**, et al. (2025). Microplastic contaminants in eggs: Identification, characterization, and potential risks. **Food Bioscience (Elsevier, Q1, IF=5.9)**
- Chakraborty, T. K., ..., **Islam, K. R.**, et al. (2025). Unveiling the hidden pollutants in the indoor environment: Focus on microplastic pollution and its related risks in the educational institutions of megacity, Bangladesh. **Environmental Pollution (Elsevier, Q1, IF=7.3)**
- Nice, M. S., ..., **Islam, K. R.**, et al. (2025). Physicochemical characterization, ecological risk assessment and spatial distribution of microplastics in Bangladesh's largest coral reef systems. *Physics and Chemistry of the Earth (Elsevier, Q1, IF=4.1)*
- Chakraborty, T. K., ..., **Islam, K. R.**, et al. (2025). Evaluating the impacts of microplastics on agricultural soil physical, chemical properties, and toxic metal availability: An emerging concern for sustainable agriculture. **Plos One (PLOS, Q1, IF=2.9)**
- Chakraborty, T. K., ..., **Islam, K. R.**, et al. (2024). Microplastics in the commercially available branded milk in Bangladesh: An emerging threat for human health. **Journal of Hazardous Materials (Elsevier, Q1, IF=12.2)**
- Chakraborty, T. K., ..., **Islam, K. R.**, et a. (2024). Application of machine learning and multivariate approaches for assessing microplastic pollution and its associated risks in the urban outdoor environment of Bangladesh. **Journal of Hazardous Materials (Elsevier, Q1, IF=13.6)**
- Netema, B. N., ..., **Islam, K. R.**, et al. (2024). Appraisal of microplastic pollution and its related risks for urban indoor environment in Bangladesh using machine learning and diverse risk evolution indices. **Environmental Pollution (Elsevier, Q1, IF=7.6).**
- Chakraborty, T. K., Rahman. S., **Islam, K. R.**, et al. (2024). Application of machine learning and statistical approaches for optimization of heavy metals (Cd^{2+} , Pb^{2+} , Cu^{2+} , and Zn^{2+}) adsorption onto carbonized char prepared from PET plastic bottle waste. **AQUA-Water Infrastructure, Ecosystems and Society (IWA, Q1, IF=3.01)**
- Chakraborty, T. K., ..., **Islam, K. R.**, et al. (2023). Adsorption of acid and basic dye from the simulated wastewater using carbonized microplastic particles synthesized from recycled polyethylene terephthalate plastic waste bottles: an integrated approach for experimental and practical applications. **AQUA-Water Infrastructure, Ecosystems and Society (IWA, Q1, IF=3.01)**
- Chakraborty, T. K., ..., **Islam, K. R.**, et al. (2023). Receptor model-based sources and risks appraisal of potentially toxic elements in the urban soils of Bangladesh. **Toxicology Reports (Elsevier, Q1, IF=4.82)**
- Chakraborty, T. K., ..., **Islam, K. R.**, et al. (2023). Application of machine learning and multivariate approaches for source apportionment and risks of hazardous elements in the cropland soils near industrial areas in Bangladesh. **Ecological Indicators (Elsevier, Q1, IF=6.9)**
- Chakraborty, T. K., Ghosh, S., Nice, M. S., **Islam, K. R.**, et al. (2023). Removal of hazardous textile dye from simulated wastewater by municipal organic solid waste charcoal using machine learning approaches: Kinetics, isotherm, and thermodynamics. **Heliyon (Elsevier, Q1, IF=3.77)**

- Chakraborty, T. K., **Islam, K. R.**, et al. (2023). Black carbon derived PET plastic bottle waste and rice straw for sorption of Acid Red 27 dye: Machine learning approaches, kinetics, isotherm and thermodynamic studies. *Plos One (PLOS, Q1, IF=3.7)*
- Asif, S., M., H., ..., **Islam, K. R.**, et al. (2025). Predictive Analysis and Optimization of Acid Red 27 Dye Adsorption on Rice Husk Charcoal Using Box-Behnken Design and Artificial Neural Networks approaches. (Status: Under review – Heliyon).
- Debnath, P. C., **Islam, M. S.**, **Islam, K. R.**, et al. (2026). Potentially Toxic Elements in Indoor Dust from Dhaka Educational Buildings. (Status: Submitted - Indoor and Built Environment).
- Rahman. M. S., **Islam, M. S.**, Debnath, P. C., **Islam, K. R.**, et al. (2026). Road Dust as a Hidden Reservoir of Heavy Metals: Source Apportionment, Spatial Distribution, and Risk Assessment across Bangladesh's Major Metropolitan Areas. (Status: Submitted - Next Sustainability).

Conference Proceedings

- Islam, K. R.**, Rahman., M. S., ... and Ghosh, G. C. (2024). Performance evaluation of poultry litter and poultry litter biochar as a promising soil amendment for influencing agricultural soil properties and toxic metal availability. (2nd International Conference on Environment, Khulna University)
- Islam, K. R.**, Nice, M. S., ... and Ghosh, G. C. (2024). Analysis of the microplastics contamination, abundance and characterization in surface water and sediment along the Kuakata sea beach, Bangladesh: an emerging issue for marine ecosystem. (2nd International Conference on Environment, Khulna University)
- Rahman, M. S., **Islam, K. R.**, Nice, M. S., ... and Chakraborty, T. K., (2023). Adsorption of heavy metals (Cd^{2+} , Cu^{2+} , Pb^{2+} , Zn^{2+}) onto graphene synthesized from pet plastic bottle waste: kinetics, isotherm and thermodynamic. (9th International Conference on Water and Flood Management, BUET)
- Netema, B. N., **Islam, K. R.**, Munna, A., Haque, M. M., Chakraborty, T. K., Zaman, S., and Ghosh, G. C. (2023). Assessment of hydro-geochemistry and groundwater suitability for drinking and 6 irrigation purposes in the Jashore district, Bangladesh using GIS, and multivariate statistical approaches. (9th International Conference on Water and Flood Management, BUET)
- Nice, M. S., **Islam, K. R.**, B., ... and Chakraborty, T. K. (2023). Microplastics as an emerging pollutant in farmland soil: an impact evaluation study on farmland soil properties and toxic metal availability. (7th International Conference on Civil Engineering for Sustainable Development, KUET)
- Nice, M. S., Netema, B. N., **Islam, K. R.**, Rahman., M. S., ... and Chakraborty, T. K. (2024). Prevalence, Distribution, and Characterization of Microplastics in Water and Sediment of Saint Martin's Island, Bangladesh: An Emerging Concern for Marine Environment. (2nd International Conference on Environment, Khulna University)
- Netema, B. N., **Islam, K. R.**, ... and Chakraborty, T. K. (2024). Preparation, characterization and performance evaluation of bio-nano composite for sustainable nutrients release: Toward sustainable agricultural practices. (International Conference on Environment, Khulna University)
- Rahman., M. S., **Islam, K. R.**, Nice, M. S., ... and Ghosh, G. C. (2024). Identification, characterization and potential risks of microplastics in the urban street dust of Bangladesh: An emerging concern for ecological and public health. (2nd International Conference on Environment, Khulna University)

Training and Certification

- Hands-on Training on **GC-MS** (Gas Chromatography and Mass Spectrometry)
Held by CSIRL, Jashore University of Science and Technology, Jashore-7408, Bangladesh
Duration: 2 days
- Hands-on Training on **FTIR** (Fourier Transform Infrared Spectroscopy)
Held by CSIRL, Jashore University of Science and Technology, Jashore-7408, Bangladesh
Duration: 2 days
- Hands on training on Field Emission Scanning Electron Microscopy (**FE-SEM**)

Held by Genome Center, Jashore University of Science and Technology, Jashore-7408, Bangladesh

Duration: 3 days

- Hands-on Training on **ICP-OES** (Inductively coupled Plasma Optical Emission Spectroscopy)

Held by Genome Center, Jashore University of Science and Technology, Jashore-7408, Bangladesh

Duration: 2 days

- Certificate program on **Climate Change and Intersectionality**.

Organized by the Department of Climate and Disaster Management, JUST as a partner institution of a project led by GRRIPP South Asia and the University of London

Duration: 2 days

Awards & Recognitions

- Vice-Chancellor's Award for Excellence (2025) in recognition of outstanding research contributions.

Language Skill

- **English:** Fluent in English
IELTS overall score: 7 (L-8.0, R-7.0, W-6.0, S-6.0)
- **Bengali:** Native

Technical skills

Instrumentation: FTIR, SEM, ICP-OES, AAS, GC-MS

Software: Microsoft Office, OriginPro, ArcGIS, Adobe Photoshop

Programming Language: R (Basic)

Extra-Curricular Activities

- Voluntary work in the biological conference ICBER in Jashore University of Science and Technology, Jashore-7408, Bangladesh
- Voluntary work in the science fair programme in Jashore University of Science and Technology, Jashore-7408, Bangladesh
- Member of Rotaract Club of Jashore University of Science and Technology, Jashore-7408, Bangladesh

References

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